Total hip arthroplasty is performed to replace the arthritic femoral head and neck. The acetabulum is reamed to accept a cup made of plastic and metal. Either part may be cemented into place or not cemented, depending upon the brand of orthopedic replacement parts used.







In a bipolar hip arthroplasty, both the acetabulum and femoral head are replaced. In a unipolar hip arthroplasty, only the femoral side is replaced.

The patient is positioned either supine or lateral decubitus. Blood loss is typically 250-750 mL, but may be much higher if a revision surgery is performed because the old components must be removed first.

Without prophylaxis, the incidence of deep vein thrombosis in hip arthroplasty is greater than 50% and 2-3% with prophylaxis.

Patients with rheumatoid arthritis may have concomitant airway difficulties, cardiac valve fibrosis, pulmonary effusions, aortic regurgitation, and pulmonary fibrosis. An ECG is always indicated in rheumatoid patients presenting for major surgery.

The nature of the presenting illness makes evaluation of exercise tolerance difficult and a dobutamine stress test or thallium imaging may be necessary to ascertain cardiac risk.

Regional anesthesia or general anesthesia is suitable for this surgery, although regional anesthesia offers the advantages of decreased risk for deep vein thrombosis and pulmonary embolus as well as decreased blood loss. A T10 level is adequate and capable of providing the muscle relaxation needed for the surgery.

Microemboli may occur during the pressurized application of methylmethacrylate cement, which can lead to hypotension, pulmonary hypertension, and hypoxia. Hypovolemia exacerbates the adverse effects of the cement application and should be avoided.

Jaffe RA, Schmiesing CA, Golianu B. Anesthesiologist's Manual of Surgical Procedures. 5th ed. Philadelphia, PA: Wolters Kluwer Health, 2014: 1018-1020, 1023-1026.

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